

ALEKHYA SOMU

✉ somualekhya2002@gmail.com ☎ 480-942-9909 in LinkedIn

EDUCATION

Arizona State University

Tempe, AZ

MS in Information Technology Project Management - GPA: 4.0/4.0

Jan 2024 – Dec 2025

- **Coursework:** Cloud Architecture, Natural Language Processing, Advanced Database Management Systems, Advanced Big Data Analytics, Information Systems Development, Data Visualization, Advanced Information Systems Security

Kallam Haranadhareddy Institute of Technology

Guntur, India

B.Tech in Computer Science and Engineering - GPA: 8.0/10.0

Aug 2019 – May 2023

- **Coursework:** Data Structures and Algorithms, Computer Networks, Database Management Systems, Operating Systems, Artificial Intelligence and Machine Learning, Deep Learning, Programming Languages, Data Science, Web Technologies

TECHNICAL SKILLS

Languages: Python(Pandas, NumPy), C, C++, Java17, HTML, CSS, JavaScript, TypeScript, Go, Prolog, XML

Frameworks: Django, Spring Boot, Bootstrap, Express.js, React, Node.js, PyTorch, TensorFlow, Keras, Flask

Tools: Git, AWS, Docker, Tableau, Power BI, Figma, Azure, MLflow, Jenkins, CI/CD pipelines, Kubernetes

Databases: MySQL, PostgreSQL, Oracle, MongoDB, NoSQL

Others: Distributed Software Development, Systems Design, Object Oriented Programming, Design Patterns, Agile Scrum

PROFESSIONAL EXPERIENCE

NL Eats Community Outreach Inc.

Remote

Software Engineer Intern

May 2025 – Nov 2025

- Developed ML and NLP models using satellite, weather, synthetic soil data, enhancing recommendation accuracy by 30%.
- Designed data-fusion pipelines combining synthetic, real-world datasets improving generalization and robustness by 35%.
- Deployed scalable model endpoints using AWS Lambda, API Gateway, S3, DynamoDB, Cognito and integrated with backend.
- Automated deployment workflows efficiently using Terraform, achieving a 40% reduction in deployment time.
- Improved model explainability, versioned datasets, and integrated frontend visualizations using React and Amplify UI.

S2 Infotech International Pvt. Ltd.

Mumbai, India

Software Engineer Intern

Jan 2023 – Oct 2023

- Using Java and JavaScript, scalable software components were designed and created. Data structures and algorithms were applied to increase performance and minimize execution time by 30%.
- Designed, implemented, and deployed features in an Agile setting while working with cross-functional teams, resulting in a 25% faster product delivery cycle.
- Debugging and code reviews were performed, high code quality and system dependability were ensured, and RESTful APIs and backend services were built and optimized.

PROJECTS

Vendor Risk & Contract Compliance Platform | React, AWS Lambda, AWS Textract, Amazon OpenSearch

- Serverless SaaS to ingest contracts, extract clauses via OCR and NLP, and provide secure reviewer workflows and dashboards.
- Automated clause classification and vendor risk scoring; clause-level search and similarity for fast compliance reviews.
- Built with React and different AWS tools like Textract, Lambda, S3, OpenSearch, Aurora, Cognito, QuickSight.

Secure Chaff-less Fuzzy Vault | Python, PyTorch, OpenCV

- Implemented two advanced deep learning models for high-accuracy face recognition and robust feature extraction.
- Developed a secure and efficient biometric authentication system using Chaff-less Fuzzy Vault technique to ensure robust template protection, effectively mitigating risks associated with biometric data exposure and unauthorized access.
- Conducted rigorous testing on LFW and VGG2 datasets, achieving 96.7% accuracy, and ensuring zero false acceptance.

Intelligent Road Maintenance Assistant | YOLOv8, Python, Fast API, React

- Designed and trained a YOLOv8-based detector to identify and segment potholes, cracks, and road defects in images.
- Implemented severity scoring and geo-clustering to prioritize repairs, and computed priority heatmaps for municipal action.
- Built FastAPI inference pipeline and React dashboard with map visualization, CSV export, and demo deployment.

CERTIFICATIONS

- Machine Learning for Natural Language Processing, AWS Academy
- Data Analytics with Python, by IIT Roorkee
- Machine Learning, by Andrew Ng, Coursera
- Cloud Architecting Badge, AWS Academy